Deep Learning to Predict Hospitalization at Triage: Integration of Structured Data and Unstructured Text

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***Abstract*— Overcrowding in Emergency Departments (ED) is considered as an international issue, which could have adverse impacts on multiple care outcomes such as the length of stay for example. Part of the solution could lie in the early prediction of the patient outcome as discharge or hospitalization. This study applies Deep Learning to this end. A large-scale dataset of about 260K ED records was provided by the Amiens-Picardy University Hospital in France. In general, our approach is based on integrating structured data with unstructured textual notes recorded at the triage stage. The key idea is to apply a multi-input of mixed data for training a classification model to predict hospitalization. In a simultaneous manner, the model training utilizes the numeric features along with textual data. On one hand, a standard Multi-Layer Perceptron (MLP) model is used with the standard set of features (i.e. numeric and categorical). On the other hand, a Convolutional Neural Network (CNN) is used to operate over the textual data. The two components of learning are conducted independently in parallel. The empirical results demonstrated that the classifier could achieve a very good accuracy with ROC-AUC≈0.83. The study is conceived to contribute to the mounting efforts of applying Natural Language Processing in the healthcare domain.**

Keywords—Emergency Department, Hospitalization, Deep Learning, NLP, CNN.

I. INTRODUCTION

Healthcare systems face fundamental challenges implied by population ageing and the increasing demand for health services. In Europe, for example, the elderly population aged 65 years or above is expected to double by 2060 [24]. More significantly, the proportion of elderly aged over 80 years is expected to triple between 2008 and 2060 [25]. Equally important, the serious implications imposed by the COVID-19 pandemic are adding enormous pressure on different aspects of the healthcare delivery (e.g. [32-34]). Viewed this way, it is clear that the healthcare field will continue to encounter emerging burdens in the years to come.

Among those challenges is the overcrowding experienced in the Emergency Department (ED). The ED crowding has become an international issue throughout the hospitals worldwide. Several studies, from different countries, emphasized that crowded EDs could lead to unintended breaches and deficiencies [16]. This could have an adverse impact on several facets related to the quality and cost of healthcare. Examples include increased boarding time [12], increased morbidity [17], and higher mortality rates [18, 19]. Accordingly, there is an imperative need for developing assistive tools that can help with the planning and management of ED services.

In this regard, Data Analytics could effectively provide part of the solution. The environment of healthcare generally involves data-rich procedures that generate vast amounts of data on a regular basis. Likewise, the operations of medical triage include multiple readings and observations, which are typically made by nurses and physicians. Machine Learning (ML) is therefore well-poised to act as an enabling artefact to help make sense of the ED data. For instance, several studies (e.g. [16] [21]) reported that the early prediction of outcomes contributed positively to reducing the length of stay (LOS) in the ED units.

This study is particularly concerned with the early prediction of patient hospitalization at the triage stage. The study makes use of a large-scale dataset collected by the ED of the Amiens-Picardy University Hospital (APUH) in France. The dataset contained about 260K ED records collected over more than four years. In essence, our approach aims at integrating structured data with unstructured textual notes taken by nurses or physicians at the ED. The potentials of Natural Language Processing (NLP) for healthcare have been largely discussed in the literature (e.g. [9-11]). However, the literature still lacks applications of NLP in the context of predicting hospital admissions at the ED.

The study attempts to make contributions in two aspects as follows. First, the study practically explores the suitability of free-text triage notes for developing ML models for predicting hospitalization at accuracy. We present a Deep Learning-based architecture that could integrate structured data along with unstructured text to this end. While in a broader context, the study is conceived to contribute to the ongoing efforts towards availing of NLP-driven methods in the healthcare domain.

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II. Related Work

Plentiful studies sought to develop methods in order to help predict hospital admissions. On one hand, the larger part of efforts developed predictive models based on structured data. On the other hand, further studies attempted to utilize unstructured data such as free-text notes made by nurses or physicians. In this context, our review aimed to investigate representative contributions from both aspects. The review is unavoidably selective rather than exhaustive.

A. Using Strucutured Data

A recent study [1] used a dataset from the National Hospital and Ambulatory Medical Care Survey (NHAMCS) [2] over the period 2007–2015. The NHAMCS dataset included over 200K records of adult ED visits. The triage information contained structured data fields such as patient age, sex, mode of arrival, and triage vital signs (e.g. temperature, respiratory rate, and oxygen saturation, etc.). A variety of ML models were applied to predict the hospitalization outcome at triage. The models included Logistic Regression, Random Forest, Gradient Boosting Classifier (GBC), and Deep Learning. The GBC and Deep Learning models could achieve the highest performance as 0.85, and 0.86 of the ROC-AUC respectively.

Another study [3] used a large-scale dataset including more than 1.2M records of ED visits. The dataset spanned over ten years in a large urban hospital in Singapore. The predictive model was trained using a set of eight variables including age group, race, postal code, day of week, time of day, triage category, mode of arrival, and fever status. Using a Logistic Regression model, they could achieve a ROC-AUC of 0.825.

In a longitudinal investigation [4], data were combined from electronic health records related to about 4.5M patients across England between 1985 to 2015. The initial set of predictors included 43 variables related to patient demographics, lifestyle factors, laboratory tests, morbidities, and previous ED admissions. The dataset was integrated with other information (e.g. marital status, prior general practice visits, and morbidities), and temporal information (e.g. time since first diagnosis). Based on a 5-fold cross-validation, the GBC and Random Forest models could achieve a ROC-AUC of 0.85 and 0.83 respectively.

While a study [5] developed a Logistic Regression model to predict hospitalization in a progressive manner. The model was trained to make predictions at three consecutive time points over the ED visit including the first 10 min, an hour later, and within 2 hours. In general, the literature is replete with contributions that applied ML with modalities in this context (e.g. [6-8]).

B. Using Unstrucutured Data

There has been a growing interest in applying NLP-driven methods, transformations, and algorithms to make use of textual data at the triage stage. For example, a study utilized the bag-of-words representation of triage free-text notes [13]. Using a dataset of over 250K ED visits, neural network models were trained to predict hospital admissions. They could achieve a promising accuracy with ROC-AUC≈0.74.

Further studies sought to analyze the suitability of free-text notes for building predictive models. To name a few, it was aimed to compare the performance of ML models with the inclusion of textual elements [12]. The comparative analysis included Logistic Regression and Neural Network models to predict hospital admission or transfer after presentation to the ED triage. The key finding of their work is that the predictive accuracy generally improved with the addition of free-text fields (e.g. reason for visit variables). Another study applied Deep Learning along with Word Embeddings using clinical narratives [14]. Likewise, it was reported that the NLP-based approach outperformed other methods.

A recent study [15] applied the Bidirectional Encoder Representations from Transformers (BERT), a state-of-the-art NLP approach. Using an extensive dataset of 1.8M free-text notes, the BERT model was applied to derive contextual embeddings. They interestingly demonstrated that such contextual embeddings could accurately map textual data of similar meaning to points of proximity in a vector space.

III. Methodology

A. Data Description

The present work used a dataset provided by the Amiens-Picardy University Hospital (APUH) in France. The dataset comprised more than 260K ED records over the period of January 2015 to June 2019. Each record was associated with a binary label representing the outcome at triage (i.e. hospitalization or discharge). The overall percentage of hospitalized patients was about 35%. The average LOS in ED was 4h23m. The average LOS of discharged patients was 4h04m, which was about 18% lower than those hospitalized. Table 1 gives summary statistics of the dataset.

The dataset included a variety of variables. Specifically, a set of 38 variables representing numeric, categorical, and textual values. Part of the numeric variables described standard vital signs of the patient such as temperature, oxygen flow, heart rate, respiratory rate, etc. While the categorical variables described the general patient characteristics such as gender, origin, family status, etc. The categorical variables were represented using data dictionaries provided by the hospital system.

In addition, the dataset contained other fields of free-text input. Specifically, a set of four textual fields described the triage-provided observations, psychiatric history, surgical history, and medical history of patients. The free- text notes were expressed in the French language by nurses or physicians at the ED. However, it is worth mentioning that the dataset suffered from blank textual fields across thousands of records. Roughly about 70K records did not include any textual information at all.

TABLE 1. SUMMARY STATISTICS.

Year	Number of Records	Hospitalization %
2015	53,465	35.6%
2016	57,963	35.4%
2017	60,155	34.1%
2018	61,794	34.0%
2019	30,615	33.1%

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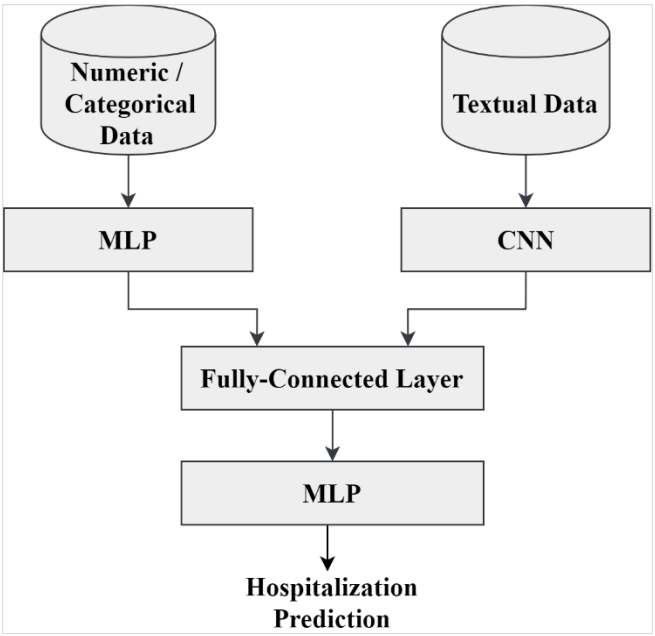


Figure 1: Approach overview.

B. Learning Appraoch

The key idea of our approach is to apply a multi-input of mixed data for training the classification model to predict hospitalization. More precisely, the model simultaneously utilized the numeric features along with the textual data within the training process. On one hand, a standard Multi-Layer Perceptron (MLP) model is used with the standard set of features (i.e. numeric and categorical). On the other hand, a Convolutional Neural Network (CNN) is used to operate over the textual data. The two components of learning are conducted independently in parallel.

Subsequently, a global feature map is produced by concatenating the outputs produced by the MLP and CNN models. The feature map is eventually fed to an MLP model, which would perform the classification task. Figure 1 sketches the key elements of our approach.

C. Data Pre-Processing

The pre-processing of data is a challenging part of the mixed-input approach, which may require different types and order of procedures. In this regard, different pipelines of pre-processing were applied on each side of the input data. This section only describes the pre-processing procedures of numeric and categorical features.

At the outset, a number of irrelevant variables were excluded from the raw dataset (e.g. Record ID, discharge date, discharge destination). The included features are given in Table 2. As it appears, the dataset eventually contained 15 numeric features, and 9 categorical features.

The dataset originally suffered from missing values in a variety of columns. The missing values of vital signs (e.g. blood pressure, temperature, etc.) were filled with routine values commonly given in the medical domain. Blank categorical values were set to default values defined by the data dictionaries.

TABLE 2. FEATURES USED FOR MACHINE LEARNING.

#	Field Name	Type
1	Arrival (Week Day /Hour)	Categorical
2	Gender	Categorical
3	Origin	Categorical
4	Arrival Modlaity	Categorical
5	Accompaniers	Categorical
6	Family Status	Categorical
7	Waiting Modality	Categorical
8	Reason for Encounter	Categorical
9	Circumstances	Categorical
10	Age	Numeric
11	Oxygen Flow	Numeric
12	Heart Rate	Numeric
13	Respiration Rate	Numeric
14	Systolic Blood Pressure	Numeric
15	Diastolic Blood Pressure	Numeric
16	Pain Scale	Numeric
17	Temperature	Numeric
18	Oxygen Saturation	Numeric
19	Capillary Blood Glucose	Numeric
20	Capillary Blood Hemoglobin	Numeric
21	Bladder volume	Numeric
22	Capillary Blood Ketones	Numeric
23	Breath Test of Alcohol	Numeric
24	Nurse Triage Scale	Numeric
25	Nurse Notes	Text
26	Psychiatric History	Text
27	Surgical History	Text
28	Medical History	Text

The min-max normalization was applied to all numeric variables, such that all values were linearly scaled to [0, 1]. While the categorical features were encoded as one-hot numeric arrays. The pre-processing procedures were applied using the Scikit-Learn [26] and Pandas [27] Python libraries.

D. Text Normalization

As it appears in Table 2, the dataset included four columns that stored textual notes. To facilitate the processing and learning, the initial step was to combine those columns into a single field. The merged text can be considered as a document associated with each patient record. Subsequently, standard procedures of text normalization were applied in order to clean and standardize the textual data into an amenable form for ML. The procedures included case conversion, removing stop words, etc.

In addition, the word stemming was applied to obtain the base form of words. Stemming helped further to reduce the degree of randomness in the text, and narrowing its content down to relatively predefined forms. The *SnowballStemmer* algorithm for the French language was used in this regard. Text normalization procedures were implemented using the NLTK library [20].

E. Tokenization

Text tokenization is the process of splitting textual data into smaller meaningful components called tokens [28]. Following the normalization process, the text was in a suitable form for tokenization. We used the Keras library [23], which provides a convenient method for tokenization.

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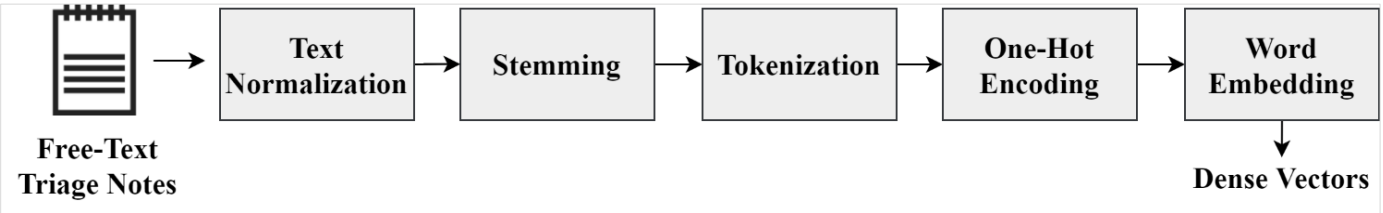


Figure 2: The pre-processing pipeline of textual data.

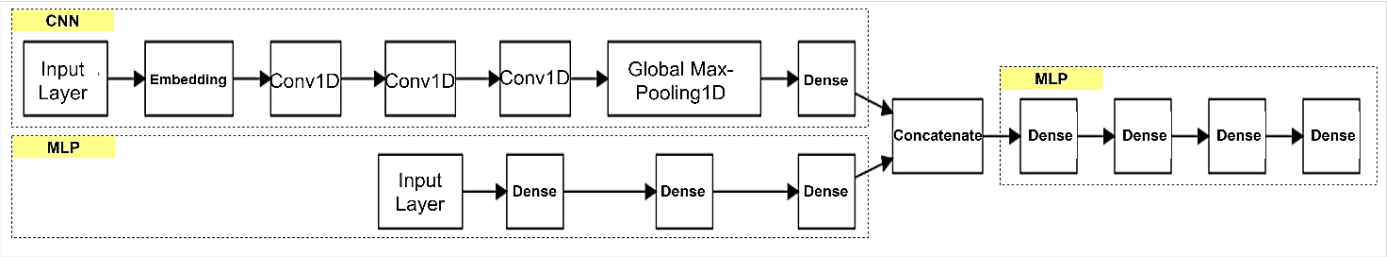


Figure 3: Model architecture.

Using the Tokenizer utility class, textual sentences could be vectorized into a list of integer values. Each integer was mapped to a value in a dictionary that encoded the entire corpus. The dictionary keys represented the vocabulary terms.

F. One-Hot Encoding

Subsequently, tokens were represented as vectors by applying the one-hot encoding. It is a simple process that produces a vector of the length of the vocabulary with an entry for each word in the corpus. In this way, each word would be given a spot in the vocabulary, where the corresponding index is set to one. Keras also provides an easy-to-use APIs for applying the one-hot encoding. As such, the sequences were eventually represented as one-hot encoded strings.

G. Word Embedding

The final step was to apply the word embedding technique, which is a vital procedure for making the text sequences tractable for ML. The one-hot encoded vectors usually suffer from high dimensionality and sparsity, which makes the learning vulnerable to the curse of dimensionality [31]. In this respect, embeddings were applied to provide dense vectors of much lower dimensions compared to the encoded representation.

Keras [23] provides a special Neural Network layer for the implementation of embeddings. The embedding layer can be basically regarded as a dictionary that maps integer indices into dense vectors. Based on integer inputs, it looks up these values in an internal dictionary, and returns the associated vectors. The embedding layer was used as the top layer of the CNN model. Figure 2 illustrates the pipeline of pre-processing procedures.

IV. Experimental Results

A. Experimental Dataset

Since our approach was genuinely based on the integration of numeric and textual data, we unavoidably had to exclude the records where no textual notes were present at all. As such, the experimental dataset included about 190K records overall. The dataset was split into train and test sets based on a 3-fold cross-validation process.

B. Model Design

The model architecture was designed to include two separate channels that could deal with the numeric and text data separately. On one hand, a standard MLP model, including three hidden layers, was used against the numeric and categorical set of features. The layers included 128, 64, and 32 neurons, respectively. The MLP design was largely determined empirically.

On the other hand, the model included another CNN-based channel that performed the feature extraction from text data. Using 1D-convolutional layers was appropriate for the case of single-dimensional sequences of data (i.e. text strings). The CNN layers were stacked in the following order:

- Three 1-D convolutional layers (filter size=16, 32, 64),
- Single (global-max) pooling layer, and
- Dense layer.

Eventually, the outputs of the CNN and MLP channels were concatenated to construct a global feature map. The feature map was used to train another MLP to perform the classification task. The MLP classifier included three hidden layers of 100, 50, and 25 neurons, respectively. Dropout layers [29] were utilized throughout the model layers to help reduce the possibility of over-fitting. Further, the batch normalization [30] technique was applied. Figure 3 sketches the model architecture.

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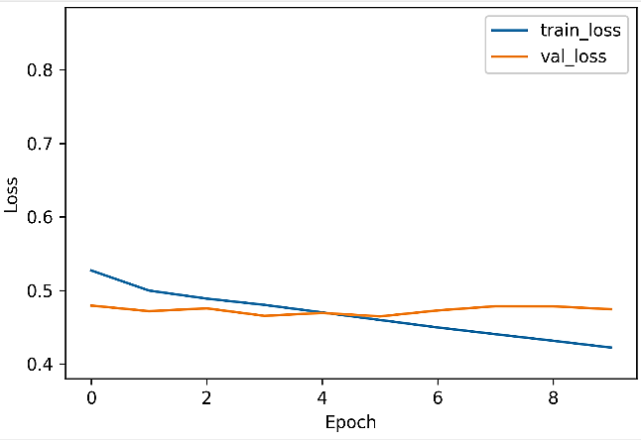


Figure 4: Model loss in train and validation.

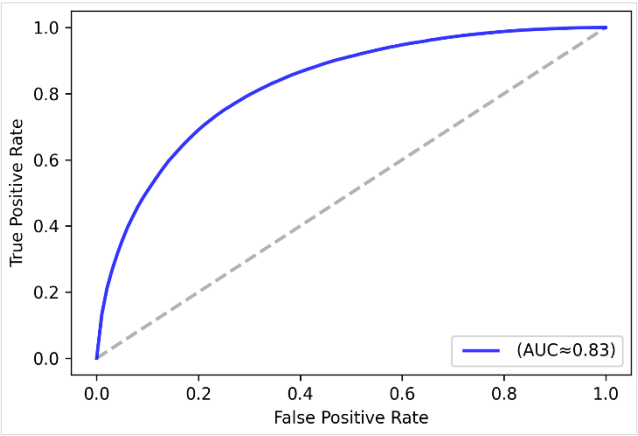


Figure 5: ROC curve.

The model was implemented using Keras [23] with the TensorFlow backend [22]. Three rounds of cross-validation were applied for evaluating the model accuracy. Figure 4 demonstrates the model loss in train and validation sets over 10 epochs, while 20% of the training set was used for model validation. Figure 5 plots the ROC curve of the model. The model could notably achieve a very good classification accuracy with ROC-AUC ≈ 0.83. More specifically, the Precision-Recall were $\approx$72% and 62%, respectively.

V. CONCLUSIONS

The overcrowding in EDs is one of the pressing challenges to healthcare services worldwide. The main objective of this paper was to apply Data Analytics to provide an early prediction of the patient hospitalization. The significance of our approach is the effective integration of structured data in tandem with unstructured free-text notes recorded at the triage stage. The model could achieve high accuracy, which encourages for employing such predictive tools to help with the care planning, and improvement of the patient journey.

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